

NATIONAL EXAM OF HIGHER NATIONAL DIPLOMA PROPOSAL – 2026 SESSION

Specialty/Option: Software Engineering

Paper: Practice of Computer

Duration: 5 hours

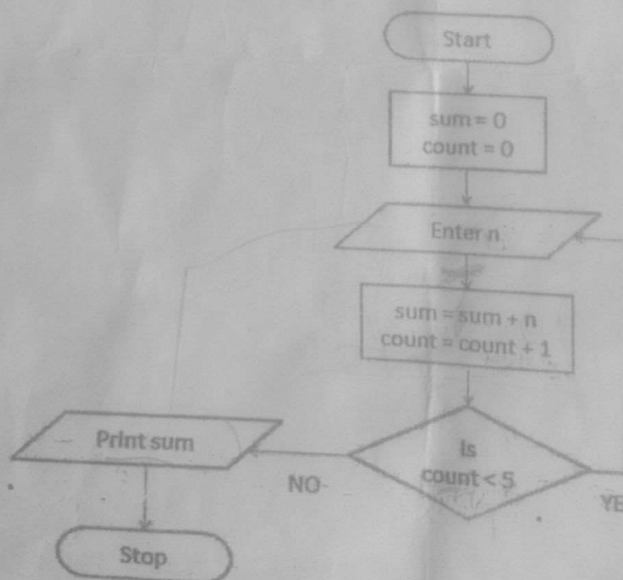
Credit Value: 08

SECTION A: ALGORITHM AND PROGRAMMING

/50 marks

Part I:

1. Define a Flowchart and explain its importance in programming. (3 marks)
2. Identify the standard symbols used for "Start/End", "Input/Output", and "Decision" in a flowchart. (3 marks)
3. Given the following flowchart:



- a. List the variables used in the flowchart and indicate their data types and functions. (3 marks)
- b. Which C structure (loop or selection) can we used to convert this flowchart to code? Justify your choice (3 marks)

- c. List two software's which can be used to write down the corresponding C code of this flowchart (3 marks)
- d. Explain what the flowchart is doing (3 marks)
- e. Modify the flowchart to accept only positive numbers as input before execution (5 marks)
- f. Using the software install in your computer, write down the corresponding java program (10 marks)

### Part 2:

A school intends to manage basic student information using Object-Oriented Programming concepts. As a student in Software Engineering, you are required to design and implement a simple system that represents a student.

### Questions:

1. Create a **class diagram** that represents a Student class. The diagram should clearly show the class name, its attributes, and their corresponding data types. (5 marks)
2. State the **software tool** that can be used to design the class diagram. (2 marks)
3. Write a program in **C++ or Java** that implements the student class with appropriate attributes such as name, matricule, age, and department. Instantiate an object of the class, assign values to its attributes, and display the student's information on the screen. (10 marks)

## SECTION B: DATABASE DEVELOPMENT AND ADMINISTRATION ( MYSQL) /20 marks

Given the following tables:

**DOCTOR**

| DOC_ID | DOC_NAME     | SPECIALIZATION |
|--------|--------------|----------------|
| D01    | Dr. Atangana | Cardiology     |
| D02    | Dr. Bello    | Pediatrics     |
| D03    | Dr. Chen     | General Med    |

**PATIENT**

| PATIENT_ID | PATIENT_NAME | ASSIGNED_DOC |
|------------|--------------|--------------|
| P99        | John Doe     | D01          |
| P88        | Mary Ngo     | D03          |
| P77        | Musa Ali     | D03          |

- Task 1:** Draw the ERD showing the relationship between Doctors and Patients. (4 marks)
- Task 2:** Install the SQL editor provided on the USB flash drive supplied to you. (3 marks)
- Task 3:** Define the schema: identify Primary Keys and the link between the two tables. (3 marks)
- Task 4:** SQL Tasks:

- a- Create a database named "HOSPITAL DB". (1 mark)
- b- Create the tables and insert the records provided. (4 marks)
- c- List all Doctors and their Specializations. (1 mark)
- d- Find all patients assigned to "Dr. Chen". (2 marks)
- e- Count how many patients are assigned to each Doctor ID. (2 marks)

**Note:** Each command should be copied and pasted into a Word document titled **SWE2026.docx**, which should be saved in a folder named **SWE2026** that you have created on your desktop.

### SECTION C: WEB DESIGN

/15 marks

Given the following web page:

## WEB DESIGN SWE HND PRACTICAL 2026 PRACTICAL

Registration Form - HND STUDENT Portal

### Register

Full Name:

Email:

Password:

Age:

Preferred Language:

☐ French ☐ English

Town:

Favorite Sport(s):

☐ Football ☐ Basketball ☐ Handball

☐ Tennis ☐ Volleyball ☐ Other

Submit Registration

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- Describe each element required to write and run a web page. (2 marks)
- List the fundamental HTML elements used in creating this webpage (3 marks)



3. Using appropriate software and HTML, JavaScript, or PHP, write the equivalent code for the web page described above.

An example of the expected output message when the user submits the form should be clearly shown. (10 marks)

☒ Registration Successful

Name: CHEN LIU  
Email: chenliu@gmail.com  
Age: 58 years  
Language: French  
Town: Mokolo  
Sport(s): Tennis, Volleyball

Welcome to Web Design SWE 2026!

#### SECTION D: NETWORKING

/15marks

A university in Buea, Cameroon intends to improve its campus communication system by setting up a computer network for two new faculties. The network administrator has been given the IP address 10.10.0.0/24 and is required to divide it into four sub-networks using a /26 subnet mask.

These four sub-networks will be used as follows:

- Faculty of Science (50 computers)
- Faculty of Arts (40 computers)
- Administration Block (20 computers)
- Library & E-Learning Center (30 computers)

You are recruited as a network technician to assist in designing and simulating this network.

#### Tasks:

- 1- List all the network components required to set up this network infrastructure (2 marks)
- 2- Explain the role of a Gateway in a computer network (1 mark)
- 3- Mention one software tool that can be used to simulate and test this network before implementation (1 mark)
- 4- Given the subnet 10.10.0.0/26, determine the IP ranges for all four subnets (2 marks)
- 5- After installing the corresponding software found in the flash provided, design and illustrate the network topology (4 marks)
- 6- After successfully designing and simulating the network using the software mentioned in the previous question, proceed to the practical implementation of the network design. (5 marks)